

Semester V (B.Tech)

Er. No.
Academic Year: 2022-23

Jaypee University of Engineering & Technology, Guna

T-1 (Odd Semester 2022)
18B14PH542 – NANO SCIENCE

Maximum Duration: 1 Hour

Maximum Marks: 15

Notes:

1. This question paper has 05 questions.
2. Write relevant answers only.
3. Do not write anything on question paper (Except your Er. No.).

	Marks
Q1. Who used the term 'Nanotechnology' for the first time? How nanotechnology was used in making Lycurgus cups by the Romans in the 4 th century?	[03]
Q2. Mathematically, prove that seven-fold rotational symmetry is not possible in crystals.	[03]
Q3. Calculate the Miller indices of the crystal plane which cuts the crystal axes at $3a$, $2b$, and $-3c$.	[03]
Q4. Show that the packing factor for the FCC unit cell is 0.74.	[03]
Q5. Write Bragg's equation for diffraction. If X-rays of wavelength $1.54 \text{ \AA}$ are incident on a crystal at an angle of 32° , then what should be the value of the spacing between the planes of atoms for the first-order diffraction?	[03]

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Semester V (B.Tech.)

Er. No.....

Academic Year: 2022-23

Jaypee University of Engineering & Technology, Guna
T-1 (Odd Semester 2022)

18B19HS599-Indian Constitution and Traditional Knowledge

Maximum duration: 1 Hour

Maximum Marks: 15

Notes:

1. This question paper has 03 questions.
2. Write relevant answers only.
3. Do not write anything on question paper (Except your Enrl No.)

- Q1.** Explicate the objectives and roles of various Acts passed during 'the Company rule' and 'the Crown rule'. Elaborate the difference between the Acts passed during 'the Company rule' and 'the Crown rule' by mentioning the key features. [05]
- Q2.** Explain in detail the key features of 'the Government of India Act of 1935'. Why this Act is considered to be an important Act? [05]
- Q3.** Expound the objectives of 'the Indian Independence Act of 1947'. Elaborate various provisions that are made in this Act. [05]

Semester V (B.Tech)

Er. No. 2
Academic Year: 2022-23**Jaypee University of Engineering & Technology, Guna**

T-1 (Odd Semester 2022)

21B14CI742 – Full Stack Development

Maximum Duration: 1 Hour

Maximum Marks: 15

Notes:

1. This question paper has 3 questions.
2. Write relevant answers only.
3. Do not write anything on question paper (Except Er. No.)

Q1.

Suppose you were to create the landing page of the website of Jaypee University of Engineering and Technology, Guna. The stakeholders want you to ensure that the webpage is suggested by the search engines when a user provides a search string or the keywords like Jaypee University of Engineering and Technology, A grade university accredited by NAAC, Admission, and best private university in Madhya Pradesh. When asked about the description of the webpage, the stakeholders provided this information: "Jaypee University of Engineering and Technology, Guna is Grade 'A' Accredited by NAAC, and NIRF-2018 Ranked University." Since the webpage contains the live enrolment status, the stakeholders also want you to keep the website auto-refreshing after every 50 seconds. You are also expected to make this webpage as per HTML5 specifications. With the above provided information, you are directed to write the HTML5 code, such that the page is SEO friendly, bookmarking friendly and the same also contains all the necessary meta information.

Marks
[05]**Q2.**

Read the following statements in context of HTML5:

[05]

- (i) The tags for definition lists have three attributes.
- (ii) The comment style `<!-- -->` supports single line commenting only.
- (iii) `<p>` is an inline container and not a block-level element.
- (iv) `<section>` and `<div>` can be used synonymously in designing a webpage.
- (v) 'alt' attribute can go with `<img>` but make a website inaccessible.

Mark each of the aforementioned statements as 'True' or 'False' and then write about 50 words for each statement about the HTML5 topics touched in that statement.

Q3.

Write an HTML code (having only table tag) to print the following table (except border). Use only rowspan and colspan attributes inside td, tr or th tags. **[05]**

PTO

A more complex table-within-a table

Outer Table	An inner table showing a variety of headings and data items				
	Inner Table				
	Corner		Head 1		
	Head 4	Head 5	Head 2	Head 3	
	A	Too Tall		Head 6	
	B		Too Wide		
			• Item 1 • Item 2		
	Click here to go to homepage	Nested table with borders	Statement 1 on first line Statement 2 on second line Statement 3 on third line		
		<table><tr><td>Little Table</td></tr></table>	Little Table		
Little Table					

Jaypee University of Engineering & Technology, Guna

T-1 (Odd Semester 2022)

18B11CI511 – Theory of Computation

Maximum Duration: 1 Hour

Maximum Marks: 15

Notes:

1. This question paper has five questions.
2. Write relevant answers only.
3. Do not write anything on question paper (Except Er. No.).
4. (Marks are indicated in square bracket.)

- | | | Marks |
|-----|--|-------|
| Q1. | Prove the given equivalence by using logical identities, mention the identities used at each step. | [03] |
| Q2. | Obtain the principal disjunctive normal form of the given logical relation :
$(\neg P \Rightarrow (\neg P \Rightarrow (\neg P \wedge Q))) \equiv P \vee Q$
$(Q \wedge \neg R \wedge \neg S) \vee (R \wedge S)$
Write proper justification of each step used in the process. | [03] |
| Q3. | Design a deterministic finite automata (DFA) to accept strings of <i>a</i> 's and <i>b</i> 's <u>except</u> those containing the substring <u><i>aab</i></u> for the input symbol <i>a</i> and <i>b</i> only. | [03] |
| Q4. | Convert the given non-deterministic finite automata (NFA) in Fig.-1 into an equivalent deterministic finite automata (DFA). | [03] |

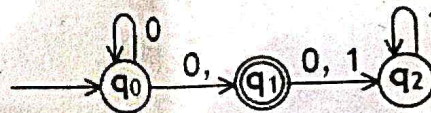


Figure 1

- | | | |
|-----|---|------|
| Q5. | Use partitioned based method to find the minimum-state FA recognizing the same language as given in Fig.-2. The input symbols are <i>a</i> and <i>b</i> only. | [03] |
|-----|---|------|

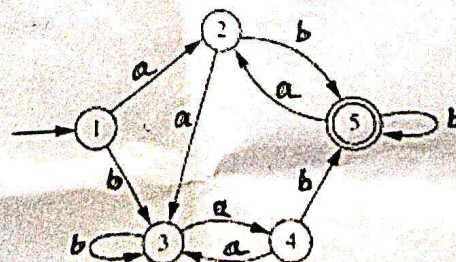


Figure 2

Semester III/V (B.Tech.)

Jaypee University of Engineering & Technology, Guna

T-1 (Odd Semester 2022)

18B11MA511- Probability Theory and Random Processes

Maximum duration: 1 Hour

Maximum Marks: 15

Notes:

1. This question paper has three questions.
2. Write relevant answers only.
3. Do not write anything on question paper (Except your Er. no.)
4. Freehand sketch of graph or diagram to be made if asked for.

Marks

[05]

Q1. State the following with suitable diagram:

- (a) Mutually exclusive events and independent events
- (b) Bayes Theorem
- (c) De Morgan's first law
- (d) Total Probability

Q2. In a certain city, 40% of people consider themselves conservatives (C), 35% consider themselves to be liberals (L), and 25% consider themselves to be independent (I). During a particular election, 45% of the conservatives voted, 40% of the liberals voted and 60% of the independents voted. Suppose that a person is randomly selected. [05]

- (a) Construct the venn diagram of the problem and find the probability that the person voted
- (b) If the person voted, find the probability that the voter is (i) Conservative, (ii) Liberal, (iii) Independent.

Q3. The distribution of the amount of gravel (in tons) sold by a particular construction supply company in a given week is a continuous random variable X with probability density function (pdf). [05]

$$f(x) = \begin{cases} \frac{3}{2}(1 - x^2), & 0 \leq x \leq 1 \\ 0, & \text{Otherwise} \end{cases}$$

Find the following:

- (a) Graph of probability density function (pdf)
- (b) Cumulative distribution function (cdf) and its graph
- (c) Probability $P(0.25 \leq x \leq 0.75)$
- (d) Expected weekly gravel sales and variance of weekly sales.